

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(b)		Attempt at masses: LHS – RHS i.e. 0.005 229 [u] or 0.004131 [u] or 0.006327 [u] (1) Attempt at conversion to MeV or mass i.e. $\times 931$ or $\times 1.66 \times 10^{-27}$ (1) Attempt at conversion to J i.e. $\times 1.6 \times 10^{-13}$ or $\times c^2$ (1) Expect 7.8×10^{-13} [J] Rearrangement and incorporating 98 % or statement ignoring 98 % i.e. $v = \sqrt{\frac{2E}{m}}$ (1) N.B. Accept 4 u or 4.0026 u or 4.0015 u 1.52×10^7 or 1.35×10^7 or 1.67×10^7 [ms ⁻¹] ecf on 98 % and one arithmetic slip (1)	1 1	1 1 1		5	5	
			Question 1 total	4	6	3	13	8	0